Grade 6
Unit 11 Assessment

Name____________________
Date_________________
Period__________

1. Select all of the expressions that are equivalent to the value of $2y - 4x$ when $x = 5$ and $y = -1$.

- ☐ -22
- ☐ -18
- ☐ $(-2) - 20$
- ☐ $(-2) + -20$
- ☐ $2(-1) + 4(5)$
- ☐ $2(-1) - 4(5)$

2. Evaluate the expression. If necessary, use the space provided to show your work.

Expression	Work
$w^3 - 5$ when $w = 2$	
Value	
3	

3. Select all of the expressions that are equivalent to the $3x + 9(2y - x)$.

- ☐ $2x + 18y$
- ☐ $18y - 6x$
- ☐ $18y - x + 3x$
- ☐ $9(2y - x) + 3x$
- ☐ $-2(3x - 8y) - 2y$

Evaluate each expression. If necessary, use the space provided to show your work.

4.

Expression	Work
$11 + 2x^2 \div (4k)$ when $x = -6$ and $k = -3$	
Value	
5	

5.

Expression	Work
$8h \div 10^3$ when $h = -500$	
Value	
-4	

6. Select all of the expressions that are equivalent to $8h + 9k - 6h + 24k$.

- ☐ $2h + 33k$
- ☐ $14h + 33k$
- ☐ $8(3k + h) + 3(3k - 2h)$
- ☐ $(8h - 6h) + (9k + 24k)$
- ☐ $3(8k + 3k) - 2(-2h)$

Evaluate each expression. If necessary, use the space provided to show your work.

7.

Expression	Work
$r^3(-4m^2 - 3(h \div 2)) \div w$ when $m = 6$, $h = 8$, $w = -2$, and $r = -1$	
Value	
-78	

8.

Expression	Work
$5(7 - 3x) + 9k$ when $x = -8$ and $k = 4$	
Value	
191	

9. Which value of x is the solution to $-6x + 22 = 8(1 - x)$?

A. -7
B. -3
C. 1
D. 2

10. Select all of the values of x that are solutions to $17 - 7x > 3(10x - 4) + 61$.

- ☐ -4
- ☐ -2
- ☐ -1
- ☐ 0
- ☐ 2
- ☐ 5

11. The American Red Cross recommends a minimum of 9 feet of water depth for head-first dives, including dives from pool decks.

a. Write an inequality to represent the depth, d, in feet, that a pool should be for someone to safely dive in head-first.

$$d \geq 9$$

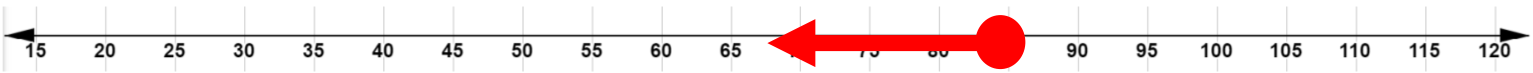
b. Represent the inequality on the number line.

12. Noise levels above 85 decibels can damage a person's hearing.

a. Write an inequality to represent the noise level, n, in decibels that will **not** damage a person's hearing.

$$n \leq 85$$

b. Represent the inequality on the number line.



13. For a fire hydrant to reach its maximum water flow, a building must be no farther than 76.25 meters (m) from the hydrant. Circle all of the distances that would result in a fire hydrant reaching its maximum water flow.

$\frac{1200}{12}$ m	76.3 m	76.2 m	$74\frac{5}{9}$ m	76.05 m
76.5 m	$76\frac{1}{8}$ m	$\frac{7620}{1000}$ m	$76\frac{12}{16}$ m	$\frac{1500}{20}$ m